AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

PHASE 7

PROJECT DOCUMENTATION PHASE

1. Introduction

Project documentation provides a complete written reference for the Auto Ticket Classification using Flow Designer system. This phase records the project requirements, design, implementation, workflow, testing, results, user guidance, and maintenance information.

The system is designed to classify incoming support tickets automatically using a flow-based process. Ticket information is collected, processed, classified according to configured rules, and routed to the appropriate destination.

2. Documentation Objectives

- Provide a complete written record of the project.
- Explain the purpose, scope, architecture, and workflow.
- Document modules, data flow, classification logic, and user interaction.
- Record testing procedures and results.
- Provide maintenance and troubleshooting guidance.
- Support future modifications and enhancements.

3. Project Overview

Auto Ticket Classification using Flow Designer is a workflow-based ticket management solution. Its purpose is to reduce manual effort in organizing support requests by applying predefined classification conditions and routing each ticket to a suitable processing path.

The project combines ticket data, workflow automation, classification rules, and routing or notification activities into a structured flow.

4. System Requirements

4.1 Hardware Requirements

- A computer or laptop with a modern processor.
- Minimum 4 GB RAM; 8 GB or more is recommended.
- Sufficient storage for project files and test data.
- A stable internet connection when cloud services are used.

4.2 Software Requirements

- Compatible operating system.
- Web browser for accessing Flow Designer.
- Flow Designer environment and required configuration.
- Database or data source for ticket information, where applicable.
- PDF reader or document editor.

5. Functional Requirements

- Accept new ticket information.
- Validate required ticket fields.
- Identify ticket type or category.
- Apply predefined classification conditions.
- Route classified tickets to the appropriate group or workflow.
- Update ticket status and classification information.
- Record processing details.
- Allow authorized users to review tickets.

6. Non-Functional Requirements

- Provide consistent classification for the same input conditions.
- Keep the workflow easy to understand and maintain.
- Handle ticket information securely.
- Provide reasonable processing performance.
- Support future expansion of categories and rules.
- Maintain clear project documentation.

7. System Architecture

The system follows a flow-based architecture. Ticket data enters through an input source. The flow validates and processes the information, applies classification logic, and directs the ticket toward the appropriate outcome. Status and classification information are then stored or communicated as required.

7.1 Major Components

- Ticket Input Module
- Validation Module
- Classification Module
- Routing Module
- Notification Module
- Data Storage Module
- Monitoring and Logging Module

8. Workflow Documentation

The workflow begins when a new ticket is received. Required information is validated, classification conditions are evaluated, and the ticket is routed to the appropriate support path.

- Step 1: Receive ticket.
- Step 2: Validate ticket information.
- Step 3: Read relevant ticket fields.
- Step 4: Apply classification conditions.
- Step 5: Assign category or priority.
- Step 6: Route the ticket.
- Step 7: Update ticket status.
- Step 8: Notify the responsible user or group.
- Step 9: Record processing information.

Figure 1: Ticket Classification Workflow

Ticket Input → Validation → Classification → Routing → Status Update → Notification → Completion.

Figure 1. Ticket Classification Workflow

9. Module Documentation

9.1 Ticket Input Module

The module receives ticket details such as subject, description, requester information, and other relevant fields.

9.2 Validation Module

The validation module checks whether required information is present and usable.

9.3 Classification Module

The classification module evaluates ticket information against predefined rules and assigns a suitable category.

9.4 Routing Module

After classification, the routing module sends the ticket to the appropriate support group, queue, or workflow stage.

9.5 Notification Module

The notification module communicates important ticket events to relevant users or groups.

9.6 Data Storage Module

This module maintains ticket information, classification results, status values, and other required records.

10. Classification Logic Documentation

The classification logic is based on predefined conditions. Keywords, ticket fields, request types, or other configured conditions may be used to determine the category.

- Read relevant ticket fields.
- Compare values with configured conditions.
- Identify the matching category.
- Assign the classification.
- Use a default or exception category when no rule matches.

11. Data Documentation

11.1 Ticket Data

- Ticket ID
- Requester details
- Subject and description
- Category and priority
- Assignment group
- Status
- Created and updated date/time

11.2 Data Validation

Required fields should be checked before classification. Validation reduces processing errors and ensures classification rules receive appropriate input.

12. User Interface Documentation

The user interface provides access to ticket information and workflow results. Users should be able to identify the ticket number, description, category, assignment information, and status.

- Ticket creation or submission screen.
- Ticket details view.
- Classification result.
- Assignment and status information.
- Workflow or activity history.

13. Implementation Documentation

The implementation consists of configuring the Flow Designer workflow, defining actions and conditions, connecting data sources, and testing the complete flow.

13.1 Flow Configuration

The flow is configured with a trigger followed by validation, classification, routing, and update actions.

13.2 Exception Handling

Exception paths handle missing information, unsupported ticket types, or processing failures.

14. Testing Documentation

Testing verifies that the workflow behaves correctly for different ticket inputs. Test cases should cover normal inputs, invalid inputs, boundary conditions, and classification conditions.

- Valid ticket with a recognized category.
- Ticket with missing required information.
- Ticket matching multiple conditions.
- Ticket with no matching classification rule.
- Successful routing and status update.
- Notification or assignment verification.

15. Test Results

Test results should record the input condition, expected output, actual output, and final status. Failed cases should be analyzed and corrected before deployment.

Sample Test Result

- Input: Ticket containing a known category keyword.
- Expected Result: Correct category is assigned.
- Actual Result: Category is assigned according to the configured rule.
- Status: Pass.

16. Error Handling and Troubleshooting

Common issues include missing ticket data, incorrect rule conditions, failed workflow actions, incorrect routing, or notification problems.

- Verify required ticket fields.
- Check the flow trigger.
- Review classification conditions.
- Verify routing configuration.

- Check notification settings.
- Review execution history and error messages.

17. Security and Access Documentation

Access to ticket information and workflow configuration should be limited to authorized users. Appropriate permissions should be used for viewing, modifying, and administering project components.

18. Maintenance Documentation

The system should be reviewed periodically to ensure classification rules, ticket categories, routing groups, and notifications remain relevant.

- Review classification rules periodically.
- Update categories when new ticket types are introduced.
- Revise outdated conditions.
- Monitor workflow execution and errors.
- Maintain updated documentation.

19. User Guide

Users can submit or review tickets through the configured interface. After submission, the workflow processes the ticket automatically.

- Enter the required ticket details.
- Submit the ticket.
- Review the assigned category and status.
- Check the assigned support group when necessary.
- Update or close the ticket according to the defined process.

20. Project Deliverables

- Complete Flow Designer workflow.
- Configured classification rules.
- Ticket and data configuration.
- Testing records and results.
- Project documentation.
- User guidance.
- Maintenance and troubleshooting information.

21. Limitations

- Classification accuracy depends on configured rules and ticket information.
- New ticket types may require additional conditions.
- Incorrect or incomplete input can affect classification.
- Platform or configuration changes may require workflow updates.

22. Future Enhancements

- Introduce more advanced classification techniques.
- Add additional ticket categories and routing rules.
- Provide improved dashboards and reporting.

- Add stronger monitoring and alert mechanisms.
- Integrate additional ticket sources.
- Improve automated handling of ambiguous tickets.

23. Conclusion

The Project Documentation Phase provides a complete reference for the Auto Ticket Classification using Flow Designer project. It records system requirements, architecture, workflow, modules, classification logic, data handling, testing, security, maintenance, and user procedures. Clear documentation supports reliable project use, easier maintenance, and future enhancement.